# DataBase Lab 04 ( In-Lab Task)


## Muhammad Ghulam Mujtaba Qureshi

## 24k-0535


# Task 1

Welcome Page | HR

Worksheet | Query Builder

```
SELECT FIRST_NAME,LAST_NAME, Salary AS Old_Salary, Salary + (Salary * 0.10) - (Salary * 0.05) AS New_Salary
FROM employees;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

| | FIRST_NAME | LAST_NAME | OLD_SALARY | NEW_SALARY |
|---|---|---|---|---|
| 1 | Steven | King | 24000 | 25200 |
| 2 | Neena | Kochhar | 17000 | 17850 |
| 3 | Lex | De Haan | 17000 | 17850 |
| 4 | Alexander | Hunold | 9000 | 9450 |
| 5 | Bruce | Ernst | 6000 | 6300 |
| 6 | David | Austin | 4800 | 5040 |
| 7 | Valli | Pataballa | 4800 | 5040 |
| 8 | Diana | Lorentz | 4200 | 4410 |
| 9 | Nancy | Greenberg | 12008 | 12608.4 |
| 10 | Daniel | Faviet | 9000 | 9450 |
| 11 | John | Chen | 8200 | 8610 |
| 12 | Ismael | Sciarra | 7700 | 8085 |
| 13 | Jose Ma... | Urman | 7800 | 8190 |
| 14 | Luis | Popp | 6900 | 7245 |
| 15 | Den | Raphaely | 11000 | 11550 |
| 16 | Alexander | Khoo | 3100 | 3255 |
| 17 | Shelli | Baida | 2900 | 3045 |
| 18 | Sigal | Tobias | 2800 | 2940 |
| 19 | Guy | Himuro | 2600 | 2730 |
| 20 | Karen | Colmenares | 2500 | 2625 |
| 21 | Matthew | Weiss | 8000 | 8400 |
| 22 | Adam | Fripp | 8200 | 8610 |
| 23 | Payam | Kaufling | 7900 | 8295 |
| 24 | Shanta | Vollman | 6500 | 6825 |
| 25 | Kevin | Mourgos | 5800 | 6090 |
| 26 | Julia | Nayer | 3200 | 3360 |
| 27 | Irene | Mikkili... | 2700 | 2835 |
| 28 | James | Landry | 2400 | 2520 |
| 29 | Steven | Markle | 2200 | 2310 |
| 30 | Laura | Bissot | 3300 | 3465 |
| 31 | Mozhe | Atkinson | 2800 | 2940 |
| 32 | James | Marlow | 2500 | 2625 |

# Task 2

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name, last_name, salary
FROM employees e JOIN departments d ON e.department_id = d.department_id
WHERE d.department_name IN ('IT', 'Finance')
AND salary BETWEEN 60000 AND 100000;
```

Query Result

SQL | All Rows Fetched: 0 in 0.02 seconds

| FIRST_NA... | LAST_NAME | SALARY |
|---|---|---|

# Task 3

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name, last_name, salary
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id = e.department_id);
```

Query Result

SQL | All Rows Fetched: 38 in 0.008 seconds

| | FIRST_NAME | LAST_NAME | SALARY |
|---|---|---|---|
| 1 | Steven | King | 24000 |
| 2 | Alexander | Hunold | 9000 |
| 3 | Bruce | Ernst | 6000 |
| 4 | Nancy | Greenberg | 12008 |
| 5 | Daniel | Faviet | 9000 |
| 6 | Den | Raphaely | 11000 |
| 7 | Matthew | Weiss | 8000 |
| 8 | Adam | Fripp | 8200 |
| 9 | Payam | Kaufling | 7900 |
| 10 | Shanta | Vollman | 6500 |
| 11 | Kevin | Mourgos | 5800 |
| 12 | Renske | Ladwig | 3600 |
| 13 | Trenna | Rajs | 3500 |
| 14 | John | Russell | 14000 |
| 15 | Karen | Partners | 13500 |
| 16 | Alberto | Errazuriz | 12000 |
| 17 | Gerald | Cambrault | 11000 |
| 18 | Eleni | Zlotkey | 10500 |
| 19 | Peter | Tucker | 10000 |
| 20 | David | Bernstein | 9500 |
| 21 | Peter | Hall | 9000 |
| 22 | Janette | King | 10000 |
| 23 | Patrick | Sully | 9500 |
| 24 | Allan | McEwen | 9000 |
| 25 | Clara | Vishney | 10500 |
| 26 | Danielle | Greene | 9500 |
| 27 | Lisa | Ozer | 11500 |
| 28 | Harrison | Bloom | 10000 |

# Task 4

Welcome Page | HR

Worksheet | Query Builder

```
SELECT *
FROM employees
ORDER BY department_id ASC, hire_date DESC;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

| | EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 200 | Jennifer | Whalen | JWHALEN | 515.123.4444 | 17-SEP-03 | AD_ASST | 4400 | (null) | 101 | 10 |
| 2 | 202 | Pat | Fay | PFAY | 603.123.6666 | 17-AUG-05 | MK_REP | 6000 | (null) | 201 | 20 |
| 3 | 201 | Michael | Hartstein | MHARTSTE | 515.123.5555 | 17-FEB-04 | MK_MAN | 13000 | (null) | 100 | 20 |
| 4 | 119 | Karen | Colmenares | KCOLMENA | 515.127.4566 | 10-AUG-07 | PU_CLERK | 2500 | (null) | 114 | 30 |
| 5 | 118 | Guy | Himuro | GHIMURO | 515.127.4565 | 15-NOV-06 | PU_CLERK | 2600 | (null) | 114 | 30 |
| 6 | 116 | Shelli | Baida | SBAIDA | 515.127.4563 | 24-DEC-05 | PU_CLERK | 2900 | (null) | 114 | 30 |
| 7 | 117 | Sigal | Tobias | STOBIAS | 515.127.4564 | 24-JUL-05 | PU_CLERK | 2800 | (null) | 114 | 30 |
| 8 | 115 | Alexander | Khoo | AKHOO | 515.127.4562 | 18-MAY-03 | PU_CLERK | 3100 | (null) | 114 | 30 |
| 9 | 114 | Den | Raphaely | DRAPHEAL | 515.127.4561 | 07-DEC-02 | PU_MAN | 11000 | (null) | 100 | 30 |
| 10 | 203 | Susan | Mavris | SMAVRIS | 515.123.7777 | 07-JUN-02 | HR_REP | 6500 | (null) | 101 | 40 |
| 11 | 128 | Steven | Markle | SMARKLE | 650.124.1434 | 08-MAR-08 | ST_CLERK | 2200 | (null) | 120 | 50 |
| 12 | 136 | Hazel | Philtanker | HPHILTAN | 650.127.1634 | 06-FEB-08 | ST_CLERK | 2200 | (null) | 122 | 50 |
| 13 | 183 | Girard | Geoni | GGEONI | 650.507.9879 | 03-FEB-08 | SH_CLERK | 2800 | (null) | 120 | 50 |
| 14 | 199 | Douglas | Grant | DGRANT | 650.507.9844 | 13-JAN-08 | SH_CLERK | 2600 | (null) | 124 | 50 |
| 15 | 191 | Randall | Perkins | RPERKINS | 650.505.4876 | 19-DEC-07 | SH_CLERK | 2500 | (null) | 122 | 50 |
| 16 | 135 | Ki | Gee | KGEE | 650.127.1734 | 12-DEC-07 | ST_CLERK | 2400 | (null) | 122 | 50 |
| 17 | 124 | Kevin | Mourgos | KMOURGOS | 650.123.5234 | 16-NOV-07 | ST_MAN | 5800 | (null) | 100 | 50 |
| 18 | 182 | Martha | Sullivan | MSULLIVA | 650.507.9878 | 21-JUN-07 | SH_CLERK | 2500 | (null) | 120 | 50 |
| 19 | 198 | Donald | OConnell | DOCONNEL | 650.507.9833 | 21-JUN-07 | SH_CLERK | 2600 | (null) | 124 | 50 |
| 20 | 132 | TJ | Olson | TJOLSON | 650.124.8234 | 10-APR-07 | ST_CLERK | 2100 | (null) | 121 | 50 |
| 21 | 195 | Vance | Jones | VJONES | 650.501.4876 | 17-MAR-07 | SH_CLERK | 2800 | (null) | 123 | 50 |
| 22 | 187 | Anthony | Cabrio | ACABRIO | 650.509.4876 | 07-FEB-07 | SH_CLERK | 3000 | (null) | 121 | 50 |
| 23 | 127 | James | Landry | JLANDRY | 650.124.1334 | 14-JAN-07 | ST_CLERK | 2400 | (null) | 120 | 50 |
| 24 | 126 | Irene | Mikkilineni | IMIKKILI | 650.124.1224 | 28-SEP-06 | ST_CLERK | 2700 | (null) | 120 | 50 |
| 25 | 134 | Michael | Rogers | MROGERS | 650.127.1834 | 26-AUG-06 | ST_CLERK | 2900 | (null) | 122 | 50 |
| 26 | 190 | Timothy | Gates | TGATES | 650.505.3876 | 11-JUL-06 | SH_CLERK | 2900 | (null) | 122 | 50 |
| 27 | 144 | Peter | Vargas | PVARGAS | 650.121.2004 | 09-JUL-06 | ST_CLERK | 2500 | (null) | 124 | 50 |
| 28 | 194 | Samuel | McCain | SMCCAIN | 650.501.3876 | 01-JUL-06 | SH_CLERK | 3200 | (null) | 123 | 50 |
| 29 | 186 | Julia | Dellinger | JDELLING | 650.509.3876 | 24-JUN-06 | SH_CLERK | 3400 | (null) | 121 | 50 |
| 30 | 197 | Kevin | Feeney | KFEENEY | 650.507.9822 | 23-MAY-06 | SH_CLERK | 3000 | (null) | 124 | 50 |
| 31 | 196 | Alana | Walsh | AWALSH | 650.507.9811 | 24-APR-06 | SH_CLERK | 3100 | (null) | 124 | 50 |
| 32 | 140 | Joshua | Patel | JPATEL | 650.121.1834 | 06-APR-06 | ST_CLERK | 2500 | (null) | 123 | 50 |

# Task 5

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name, last_name, hire_date,
ROUND((SYSDATE - hire_date) / 365, 2) AS experience_years
FROM employees;
```

Query Result

SQL | Fetched 50 rows in 0.002 seconds

| | FIRST_NAME | LAST_NAME | HIRE_DATE | EXPERIENCE_YEARS |
|---|---|---|---|---|
| 1 | Steven | King | 17-JUN-03 | 22.67 |
| 2 | Neena | Kochhar | 21-SEP-05 | 20.4 |
| 3 | Lex | De Haan | 13-JAN-01 | 25.09 |
| 4 | Alexander | Hunold | 03-JAN-06 | 20.12 |
| 5 | Bruce | Ernst | 21-MAY-07 | 18.74 |
| 6 | David | Austin | 25-JUN-05 | 20.64 |
| 7 | Valli | Pataballa | 05-FEB-06 | 20.03 |
| 8 | Diana | Lorentz | 07-FEB-07 | 19.02 |
| 9 | Nancy | Greenberg | 17-AUG-02 | 23.5 |
| 10 | Daniel | Faviet | 16-AUG-02 | 23.5 |
| 11 | John | Chen | 28-SEP-05 | 20.38 |
| 12 | Ismael | Sciarra | 30-SEP-05 | 20.38 |
| 13 | Jose Manuel | Urman | 07-MAR-06 | 19.94 |
| 14 | Luis | Popp | 07-DEC-07 | 18.19 |
| 15 | Den | Raphaely | 07-DEC-02 | 23.19 |
| 16 | Alexander | Khoo | 18-MAY-03 | 22.75 |
| 17 | Shelli | Baida | 24-DEC-05 | 20.14 |
| 18 | Sigal | Tobias | 24-JUL-05 | 20.56 |
| 19 | Guy | Himuro | 15-NOV-06 | 19.25 |
| 20 | Karen | Colmenares | 10-AUG-07 | 18.52 |
| 21 | Matthew | Weiss | 18-JUL-04 | 21.58 |
| 22 | Adam | Fripp | 10-APR-05 | 20.85 |
| 23 | Payam | Kaufling | 01-MAY-03 | 22.8 |
| 24 | Shanta | Vollman | 10-OCT-05 | 20.35 |
| 25 | Kevin | Mourgos | 16-NOV-07 | 18.25 |
| 26 | Julia | Nayer | 16-JUL-05 | 20.58 |
| 27 | Irene | Mikkilineni | 28-SEP-06 | 19.38 |
| 28 | James | Landry | 14-JAN-07 | 19.09 |
| 29 | Steven | Markle | 08-MAR-08 | 17.94 |
| 30 | Laura | Bissot | 20-AUG-05 | 20.49 |

# Task 6

Welcome Page | HR

Worksheet | Query Builder

```
SELECT *
FROM (SELECT first_name, last_name, department_id, salary,
ROW_NUMBER() OVER (PARTITION BY department_id ORDER BY salary DESC) rn
FROM employees)
WHERE rn <= 5;
```

Query Result

SQL | All Rows Fetched: 36 in 0.017 seconds

| | FIRST_NAME | LAST_NAME | DEPARTMENT_ID | SALARY | RN |
|---|---|---|---|---|---|
| 1 | Jennifer | Whalen | 10 | 4400 | 1 |
| 2 | Michael | Hartstein | 20 | 13000 | 1 |
| 3 | Pat | Fay | 20 | 6000 | 2 |
| 4 | Den | Raphaely | 30 | 11000 | 1 |
| 5 | Alexander | Khoo | 30 | 3100 | 2 |
| 6 | Shelli | Baida | 30 | 2900 | 3 |
| 7 | Sigal | Tobias | 30 | 2800 | 4 |
| 8 | Guy | Himuro | 30 | 2600 | 5 |
| 9 | Susan | Mavris | 40 | 6500 | 1 |
| 10 | Adam | Fripp | 50 | 8200 | 1 |
| 11 | Matthew | Weiss | 50 | 8000 | 2 |
| 12 | Payam | Kaufling | 50 | 7900 | 3 |
| 13 | Shanta | Vollman | 50 | 6500 | 4 |
| 14 | Kevin | Mourgos | 50 | 5800 | 5 |
| 15 | Alexander | Hunold | 60 | 9000 | 1 |
| 16 | Bruce | Ernst | 60 | 6000 | 2 |
| 17 | David | Austin | 60 | 4800 | 3 |
| 18 | Valli | Pataballa | 60 | 4800 | 4 |
| 19 | Diana | Lorentz | 60 | 4200 | 5 |
| 20 | Hermann | Baer | 70 | 10000 | 1 |
| 21 | John | Russell | 80 | 14000 | 1 |
| 22 | Karen | Partners | 80 | 13500 | 2 |
| 23 | Alberto | Errazuriz | 80 | 12000 | 3 |
| 24 | Lisa | Ozer | 80 | 11500 | 4 |
| 25 | Gerald | Cambrault | 80 | 11000 | 5 |
| 26 | Steven | King | 90 | 24000 | 1 |
| 27 | Neena | Kochhar | 90 | 17000 | 2 |
| 28 | Lex | De Haan | 90 | 17000 | 3 |
| 29 | Nancy | Greenberg | 100 | 12008 | 1 |
| 30 | Daniel | Faviet | 100 | 9000 | 2 |
| 31 | John | Chen | 100 | 8200 | 3 |

# Task 7

Welcome Page | HR

Worksheet | Query Builder

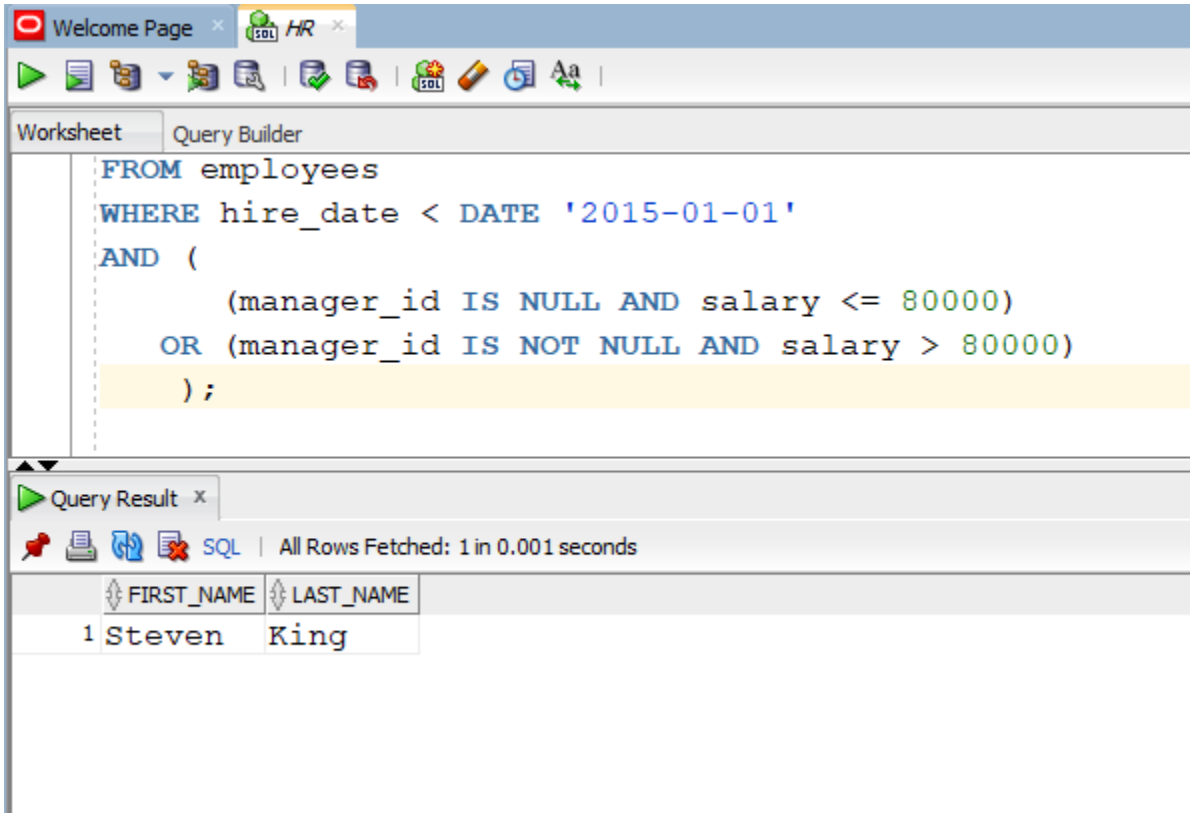
```
SELECT CONCAT(first_name, CONCAT(' ', last_name)) AS employee_name
FROM employees;
```

Query Result

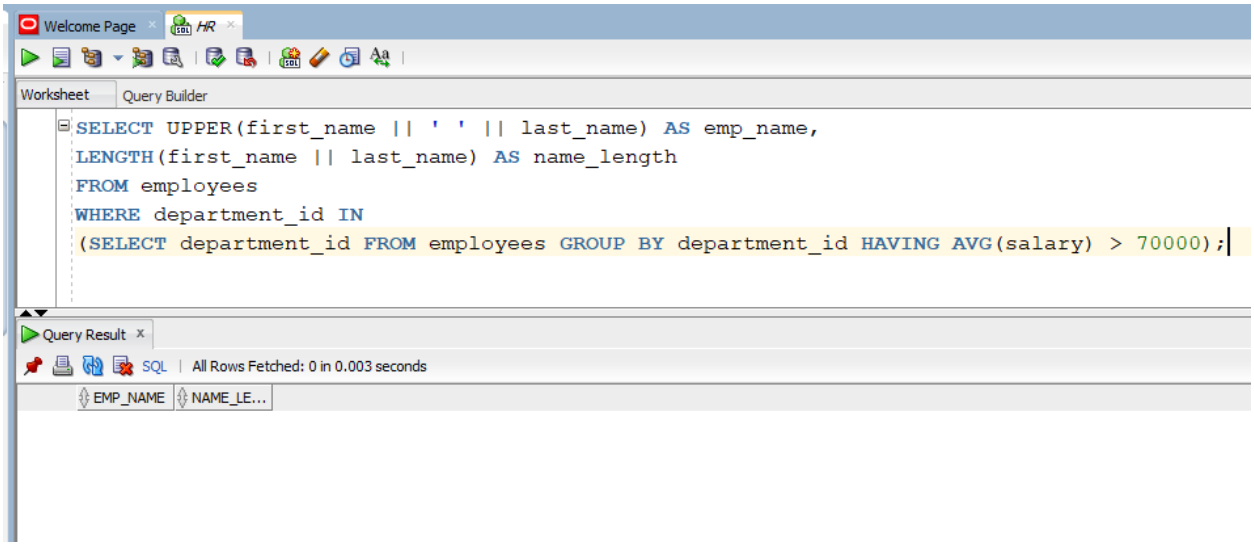
SQL | Fetched 50 rows in 0.014 seconds

| | EMPLOYEE_NAME |
|---|---|
| 1 | Ellen Abel |
| 2 | Sundar Ande |
| 3 | Mozhe Atkinson |
| 4 | David Austin |
| 5 | Hermann Baer |
| 6 | Shelli Baida |
| 7 | Amit Banda |
| 8 | Elizabeth Bates |
| 9 | Sarah Bell |
| 10 | David Bernstein |
| 11 | Laura Bissot |
| 12 | Harrison Bloom |
| 13 | Alexis Bull |
| 14 | Anthony Cabrio |
| 15 | Gerald Cambrault |
| 16 | Nanette Cambrault |
| 17 | John Chen |
| 18 | Kelly Chung |
| 19 | Karen Colmenares |
| 20 | Curtis Davies |
| 21 | Lex De Haan |
| 22 | Julia Dellinger |
| 23 | Jennifer Dilly |
| 24 | Louise Doran |
| 25 | Bruce Ernst |
| 26 | Alberto Errazuriz |
| 27 | Britney Everett |
| 28 | Daniel Faviet |
| 29 | Pat Fay |
| 30 | Kevin Feeney |
| 31 | Jean Fleaur |
| 32 | Tayler Fox |
| 33 | Adam Fripp |

# Task 8


Welcome Page
HR
Worksheet
Query Builder
FROM employees
WHERE hire_date < DATE '2015-01-01'
AND (
(manager_id IS NULL AND salary <= 80000)
OR (manager_id IS NOT NULL AND salary > 80000)
);
Query Result
SQL
All Rows Fetched: 1 in 0.001 seconds
FIRST_NAME
LAST_NAME
1 Steven King

# Task 9


Welcome Page
HR
Worksheet
Query Builder
SELECT UPPER(first_name || ' ' || last_name) AS emp_name,
LENGTH(first_name || last_name) AS name_length
FROM employees
WHERE department_id IN
(SELECT department_id FROM employees GROUP BY department_id HAVING AVG(salary) > 70000);
Query Result
SQL
All Rows Fetched: 0 in 0.003 seconds
EMP_NAME
NAME_LE...

# Task 10

Welcome Page | HR

Worksheet | Query Builder

```
SELECT first_name, last_name,
salary + (salary * 0.10) AS total_income
FROM employees;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

| | FIRST_NAME | LAST_NAME | TOTAL_INCOME |
|---|---|---|---|
| 1 | Steven | King | 26400 |
| 2 | Neena | Kochhar | 18700 |
| 3 | Lex | De Haan | 18700 |
| 4 | Alexander | Hunold | 9900 |
| 5 | Bruce | Ernst | 6600 |
| 6 | David | Austin | 5280 |
| 7 | Valli | Pataballa | 5280 |
| 8 | Diana | Lorentz | 4620 |
| 9 | Nancy | Greenberg | 13208.8 |
| 10 | Daniel | Faviet | 9900 |
| 11 | John | Chen | 9020 |
| 12 | Ismael | Sciarra | 8470 |
| 13 | Jose Manuel | Urman | 8580 |
| 14 | Luis | Popp | 7590 |
| 15 | Den | Raphaely | 12100 |
| 16 | Alexander | Khoo | 3410 |
| 17 | Shelli | Baida | 3190 |
| 18 | Sigal | Tobias | 3080 |
| 19 | Guy | Himuro | 2860 |
| 20 | Karen | Colmenares | 2750 |
| 21 | Matthew | Weiss | 8800 |
| 22 | Adam | Fripp | 9020 |
| 23 | Payam | Kaufling | 8690 |
| 24 | Shanta | Vollman | 7150 |
| 25 | Kevin | Mourgos | 6380 |
| 26 | Julia | Nayer | 3520 |
| 27 | Irene | Mikkilineni | 2970 |
| 28 | James | Landry | 2640 |
| 29 | Steven | Markle | 2420 |
| 30 | Laura | Bissot | 3630 |
| 31 | Mozhe | Atkinson | 3080 |
| 32 | James | Marlow | 2750 |
| 33 | TJ | Olson | 2310 |
| 34 | Jason | Mallin | 3630 |